Group 6
Cardiac Muscle Tissue Engineering Cryogel Scaffold for Dilated Cardiomyopathy

Phase 3 Report

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ENGS164: Tissue Engineering

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1. Background

The total prevalent cases of dilated cardiomyopathy (DCM) across the United States, Germany, France, Italy, Spain, the United Kingdom, and Japan were approximately 3,05 million in 2023. Given that DCM is a chronic condition with noticeable symptoms, a 1:1 ratio of prevalence and diagnosis is assumed[1]. The United States accounted for the largest share, representing about 51.8% of the diagnosed cases[1]. DCM is mostly prevalent in males with an approximate male-to-female ratio of 2:1[2]. There is no cure for DCM, and existing therapies do not address the underlying structural and functional damage to the heart. Therefore, there is a clear need to develop cardiac muscle scaffolds to prevent and recover from the various diseases that damage the heart.

2. Proposed Scaffold

2.1. Materials & Fabrication

The proposed injectable cryogel will be fabricated from a hyaluronic acid-gelatin (HA-co-gelatin) composite, seeded with titanium carbide (Ti2C) nanoparticles, and crosslinked with dopamine-N′, N′-methylene- bisacrylamide (DOPA-MBA). Injected into the walls of the dilated left ventricle (LV), this scaffold can increase the tissue volume (which decreases LV volume and the strain put on the heart muscle) and augment the mechanical strength of the weakened cardiac muscle. In addition, the cryogel provides the optimal environment for cardiomyocyte ingrowth and proper function. Thus, this cryogel offers both short-term and long-term treatment for DCM.

Hyaluronic acid (HA) hydrogels are flexible and have good long-term water retention. Natively found in the human body, HA is non-immunogenic and non-thrombogenic.[3,4] However, HA has been shown to be ineffective at promoting cell adhesion, which is vital for the long-term therapeutic effect of the desired scaffold.[5,6] To address this shortcoming, gelatin will also be incorporated to form a composite scaffold. Gelatin, another natural material, is a collagen derivative possessing integrin-binding domains that encourage cell adhesion.[7,8] The 2 components will be combined in an 8:2 ratio for optimal cell viability.

DOPA-MBA will be used to link the polymer scaffold. In addition to being an effective cross-linker, DOPA-MBA has been shown to increase cell adhesion and retention.[9,10] To further optimize the scaffold's ability to support cardiomyocyte function, titanium carbide (Ti_2C-8) powder will be dispersed into the polymer mixture before copolymerization. Not only would this ensure that the scaffold can function in tandem with the native tissue of the heart, but this would also provide the incorporated cardiomyocytes with the electrical stimulation required for proper development.[11,12]

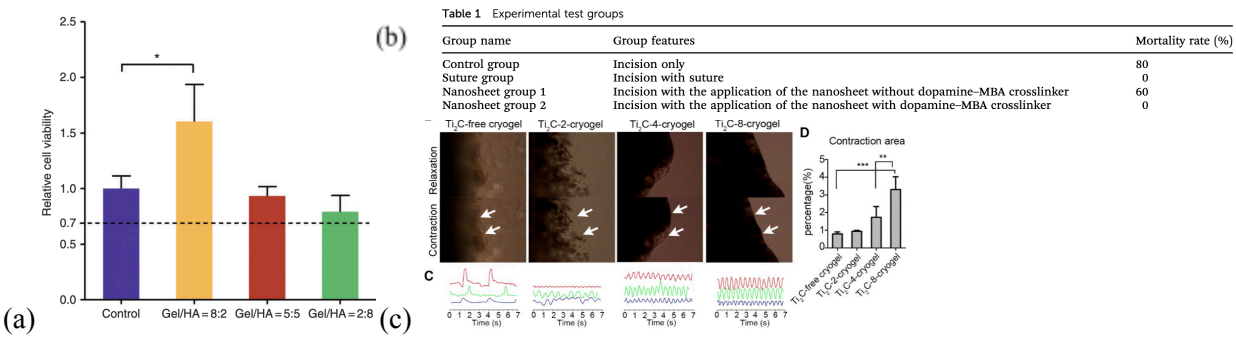
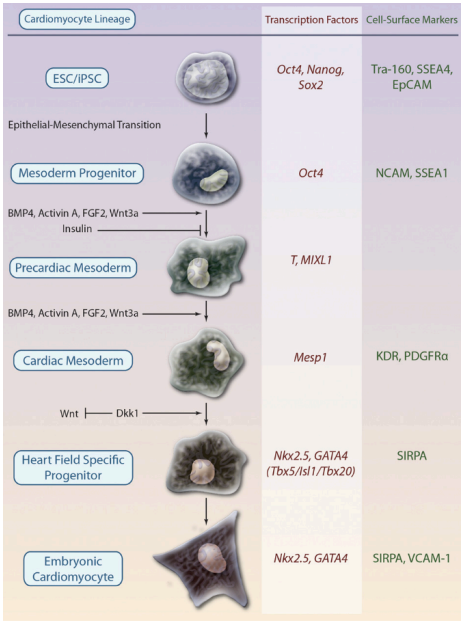


Figure 1. Figures from previous literature informing the design of the proposed scaffold. (a) 8:2 ratio of gelatin:HA yields greatest cell viability when compared to other ratios.[7] (b) DOPA-MBA results in low mortality rate when used in a nanosheet treatment for incisions, comparable to gold standard treatment.[9] (c) Ti_2C-8-seeded cryogels best restore contraction in muscle tissue when compared to other Ti_2C molecules.[12]

2.2. Cell Culture



2.2.1 iPSCs

We designed a brief procedure of differentiating iPSCs into myogenic progenitor cells, following a stepwise process that mimics early embryonic development, beginning with mesoderm induction, cardiac lineage commitment, and culminating in cardiomyocyte maturation. Initially, iPSCs are cultured in a defined differentiation medium containing Activin A and Bone Morphogenetic Protein 4 (BMP4), which play crucial roles in inducing mesoderm formation.[13] Once the mesodermal lineage is established, the differentiation protocol moves forward toward cardiac progenitor specification, where exposure to Wnt inhibitors such as IWP2 or XAV939 suppresses excessive Wnt signaling, a necessary step for commitment to the cardiac lineage.[14] See Figure 2 for the role of these growth factors.

Figure 2. Growth factors are deployed at certain stages to guide stem cell differentiation into the desired cardiomyocytes that will be used in the cryogel scaffold.[13]

2.2.3 MSCs

Similar guided differentiation of mesenchymal stromal cells (MSCs) could also be achieved by adding inducing agents during cultivation. According to Farag et al., it involves the use of 5-Azacytidine (5-AzaC), a DNA demethylation agent that activates cardiac-specific gene expression and promotes the early stages of differentiation.[15] In addition, exposure to basic fibroblast growth factor (bFGF) and transforming growth factor-β (TGF-β) enhances cardiac lineage commitment by modulating intracellular signaling pathways critical for cardio myogenesis.[16]

2.2.3 CPCs

Besides, we also intend to use Cardiac Progenitor Cells (CPCs) and Mesenchymal Stromal Cells (MSCs). CPCs support the growth of cardiomyocytes and create endothelial cells to support the structure and the vasculature necessary for repair.[17] They are a great supplement to iPSC-CMs because they are less differentiated, can proliferate, and secrete growth factors such as Hepatocyte Growth Factor (HGF), Insulin-like Growth Factor 1 (IGF-1), and Basic Fibroblast Growth Factor (bFGF) as well as cytokines that enhance cell survival. MSCs can support the growth of these new cardiomyocytes by preserving myocardial contractility, promoting angiogenesis, and reducing scar tissue formation.[18,19] CPCs can be obtained allogenically or autologously, and can also be made through iPSCs. Since we already are culturing iPSCs, it is simpler to use a genetic differentiation protocol to differentiate these cells into CPCs for the scaffold.[20] We will add 3 x 10^4 cells per cm^3 CPCs to the scaffold.[21]

2.3. Cell Seeding & Tissue Growth

Following guided cell differentiation, the cardiomyocytes could be seeded onto the cryogel scaffold, undergo a maturation phase to acquire the contractile, metabolic, and electrophysiological properties of fully functional heart cells. Additionally, we are also considering applying electrical stimulation and mechanical stretching conditions to simulate the electromechanical environment of the heart, which has been reported for improving sarcomere organization and contractile function.[22]

The myogenic cell and cryogel scaffolds co-culture methods are briefly designed as below. Firstly, the cells are maintained in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% Fetal Bovine Serum (FBS), 100 µg/mL penicillin, 100 µg/mL streptomycin, and 1% L-glutamine, under controlled incubation conditions of 37°C with 5% CO_2 to sustain proliferation.[23] The cryogel scaffolds are then sterilized with 70% ethanol for 30 minutes, washed in phosphate-buffered saline (PBS), and mechanically compressed on sterile gauze to remove trapped air and enhance cell infiltration.[24] After scaffold preparation, a concentrated cell suspension (2×10^5 cells in 10 µL) is added dropwise onto each cryogel, ensuring uniform distribution, followed by an incubation period at 37°C for 4 hours to facilitate initial attachment before transferring the constructs to fresh medium.[25] Long-term culture is maintained by replacing the medium every 48 hours, while myogenic differentiation is induced using 5-Azacytidine (10 µM), which acts as an epigenetic modulator to further promote cardiac and skeletal muscle gene expression. Insulin-like Growth Factor-1 (IGF-1) and Fibroblast Growth Factor-2 (FGF-2) are supplemented to support cell survival and differentiation.[25]

2.4. Tissue Growth Assessment

For histological analysis, the cell-scaffold co-culture system will be examined after 3 days of culture to confirm cell spreading and tissue formation. Scanning electron microscopy (SEM) and confocal microscopy will be used to analyze cell attachment and cytoskeletal organization. Furthermore, immunofluorescence staining will be conducted to detect focal adhesion proteins such as Vinculin and Integrin β1, which are key regulators of cell-extracellular matrix interactions, as illustrated in Figure 3.

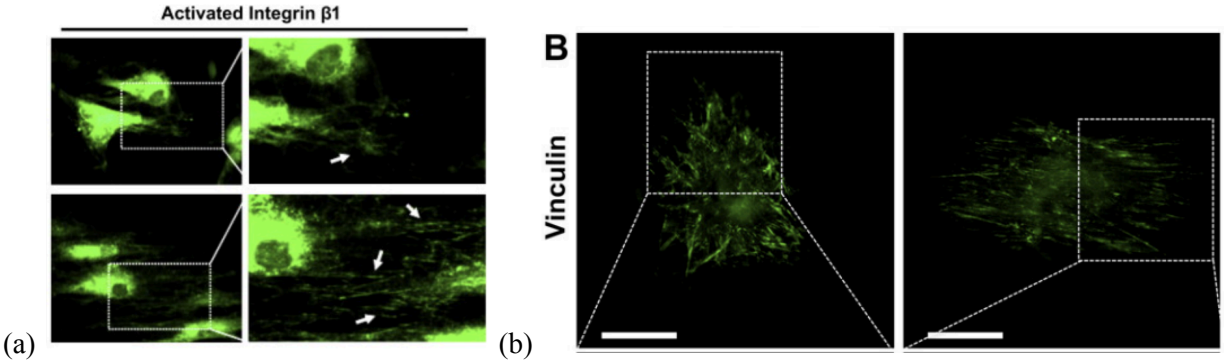


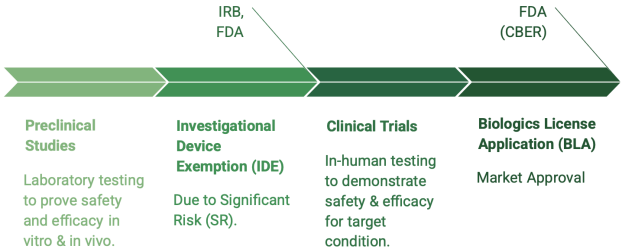
Figure 3. Sample images depicting the use of immunofluorescence staining to detect focal adhesion proteins Integrin ß1 and vinculin, which will validate the occurrence of cell-ECM interactions.

To confirm the cardiomyogenic differentiation of the seeded cells, quantitative polymerase chain reaction (qPCR) and Western blotting will be performed to detect key cardiac-specific markers. qPCR analysis will measure the expression of Nkx2.5, cardiac troponin T (cTnT), and Myocyte Enhancer Factor 2C (Mef2C), which are critical regulators of myocardial differentiation.[26] Western blotting will further validate these findings by detecting protein expression levels, confirming whether cells have successfully differentiated into functional cardiomyocytes.[27] Increased expression of these markers over time will indicate successful differentiation, while low expression may suggest that additional biochemical or mechanical cues are needed for optimal myocardial maturation.

3. FDA Approval & Clinical Trials

3.1 Combination Device

Because this proposed product fuses biologic and device components, it is considered a "combination device" that falls under Type 5: Device Coated or Otherwise Combined with Biologic.[28,29] To ensure that the proper regulatory approval pathway is pursued, the first step will be to establish contact with the FDA's Office of Combination Products (OCP). Submitting a Request for Designation (RFD) to the OCP will yield a binding classification of the product's primary mode of action (PMOA).[29] Combination products are classified by their primary mode of action (PMOA), defined as "the single mode of action of a combination that provides the most important therapeutic action of the combination product."[2] This will determine which lead center should be worked with during the investigation process, as well as the eventual marketing application.



For the combination product in question, it is possible to have either biologic or device PMOA. Because there is precedence for FDA approval of cell-seeded scaffolds as biologic PMOA, this will be used to guide the following regulatory approval plan.[30] A summary of the regulatory approval plan is presented in Figure 4, to the left.

Figure 4. Regulatory approval plan. Data from preclinical studies (phase 0) will be presented in the application for an IDE. Following IDE receipt, the scaffold will move into clinical trials to demonstrate safety and efficacy as a DCM treatment. Data collected from the clinical trials will be used to apply for a BLA.

3.2 Clinical Study Design

The nature of this combination product poses Significant Risk (SR). Therefore, the approval process for this product will require an Investigational Device Exemption (IDE), which must be approved by both the Institutional Review Board (IRB) and the FDA.[31] In addition to the preclinical testing methods and results contained in the Phase 2 report, the IDE application will detail the investigational plan of the proposed clinical study.

In order to commence clinical trials, it is necessary to provide rigorous documentation for the safety and efficacy of the product *in vitro* and *in vivo*. Many of these results will come from the experiments detailed in preclinical testing (phase 0, Section 2.4). If the submitted data is cleared, then phases I-IV of the clinical trial can begin. Phases I-III of the proposed clinical trial are outlined in the following table.

Phase (Duration)	Participants	Data Collected	To demonstrate…
I (6 M)	10, no DCM	• Patient experiences • Any adverse side effects • Heart imaging	General safety and efficacy in humans
II (18 M)	50, with DCM DCM diagnosis according to WHO criteria	• Patient experiences • Any adverse side effects • Heart imaging • Heart rate recording	Ability to decrease symptoms of and/or reverse DCM more effectively than other therapeutic options

III (18 M)	200, with DCM	● Patient experiences ● Any adverse side effects ● Heart imaging ● Heart rate recording	Safety and efficacy in various patient populations; Widespread ability to decrease symptoms of and/or reverse DCM
IV (4 Y)	2000, with DCM	● Patient experiences ● Any adverse side effects ● Heart imaging ● Heart rate recording	Long-term safety and efficacy in large, diverse patient population; Widespread ability to decrease symptoms of and/or reverse DCM

Table 1. Data collected in each phase of the proposed clinical trial, as well as the enrollment size and length of each phase.

3.2.1 Inclusion & Exclusion Criteria

The inclusion and exclusion criteria is outlined in the following tables, along with justifications. Many of these criteria have precedence in previous or ongoing clinical trials for DCM treatments.[32,33]

Ages 36-80	DCM typically develops in adulthood, with the most significant effects manifesting in older/elderly adults.
Males and non-pregnant, non-lactating females	While DCM affects men at higher rates than women, it still does affect both sexes.[34] Pregnant and lactating women are excluded to prevent complications with childbearing.
Receiving appropriate clinical standard of care heart failure therapy for > 30 days prior to screening	Patients are as healthy as possible given their DCM diagnosis.
Life expectancy > 12 months	Patients will be able to participate in the entire duration of phase II or III.
Not a candidate for reasonable, conventional revascularization procedures	Patient is not sacrificing more beneficial care for this experimental treatment.

Table 2. Inclusion criteria for clinical trial participants, with justifications.

Cardiac Comorbidities ● Severe primary valvular heart disease ● Myocardial infarction, stroke or transient ischemic attack < 3 months prior ● Other major cardiac procedure < 30 days prior	Recent/present history of other cardiac health conditions may complicate the DCM-focused assessment of the scaffold.
Left ventricular wall issues ● LV thrombus ● LV wall unsuitable for transendocardial injections	Because scaffold will be injected into the LV wall, it is vital to ensure that the LV wall is suitably healthy.
Other comorbidities	Other health issues may complicate the

● Blood clotting disorders ● Diabetic subjects ● Body mass index (BMI) ≥ 40 kg/m2 ● Chronic kidney disease (CKD) requiring dialysis ● Abnormal lab values at screening	DCM-focused assessment of the scaffold.

Table 3. Exclusion criteria for clinical trial participants, with justifications.

3.3. Market Approval

If the combination product was indeed designated a biologic PMOA, then a BLA would need to be submitted to attain market approval for this scaffold. Information such as the pre-clinical and clinical results, manufacturing process, and labeling would be required for submission to CBER.

4. Economics

Current therapies provide limited options, creating demand for regenerative approaches to address the underlying pathology rather than merely managing symptoms. The injectable cryogel scaffold would initially target severe DCM cases, approximately 25-30% of the patient population,[35] where the cost-benefit ratio is most favorable, allowing for premium pricing during market entry.

4.1. Materials Costs

The production of the scaffold involves multiple cost-intensive components, including raw materials, fabrication processes, and regulatory compliance. All product prices were sourced from Sigma Aldrich[36].

Material	CAS/Product Number	Size	Price	Unit Price
Hyaluronic acid (HA)	9004-61-9	1 g	$123.00	$0.123/mg
Gelatin	900622	1 g	$220.00	$0.22/mg
Titanium carbide	12070-08-5	25 g	$201.40	$0.008056/mg
Dopamine hydrochloride (for DOPA-MBA)	62-31-7	5 g	$44.00	$0.0088/mg
N,N′-Methylenebisacrylamide (for DOPA-MBA)	110-26-9	25 g	$65.30	$0.002612/mg
Human EGF recombinant protein	E9644	0.2 mg	$260.00	$1,300/mg
Human VEGF recombinant protein	01-185	10 µg	$481.00	$48,100.00/mg
Acetic acid (glacial)	A6283	250ML	$166.00	$6.64/mL

Table 4. Materials required for scaffold, and price.

The cryogel incorporates three different cell types. iPSC-derived cardiomyocytes are extremely costly to produce, requiring specialized equipment, reagents, and skilled technicians, approximately $10,000-25,000 per treatment.[37] Cardiac progenitor cells have similar production costs to iPSC-CMs. Mesenchymal stromal cells are more affordable at roughly $3,000 per treatment.[37] The manufacturing process requires specialized cryo-polymerization equipment with precise temperature control, clean room facilities for cell culture and scaffold preparation, quality control testing for mechanical properties, biocompatibility, and cellular viability, and GMP-compliant manufacturing facilities for clinical use.

4.2. Go-to-Market Strategy

The go-to-market strategy for the cryogel scaffold will follow a focused, phased approach. Initial market entry will target severe DCM patients who have failed optimal medical therapy and are either ineligible for or awaiting heart transplantation, positioning the therapy as a cost-effective alternative at $100,000 per treatment compared to more expensive options.[38,39] As manufacturing efficiencies reduce costs and clinical evidence accumulates, we will expand to moderate and mild DCM patients, followed by international markets and potential label expansions to other cardiomyopathies. Figure 5 depicts entry into the DCM market over time, with an initial entry into the severe market.

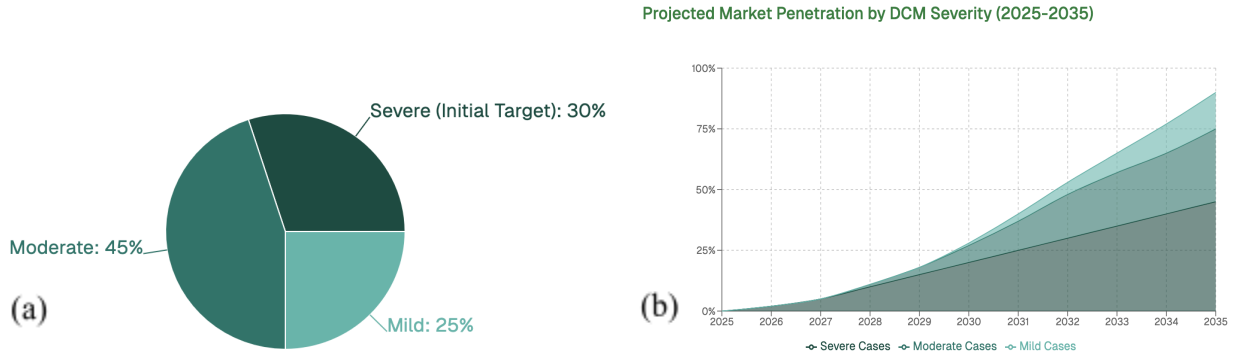


Figure 5. (a) This scaffold will enter the market as a treatment for severe DCM, which represents 30% of current DCM cases. (b) Entry into moderate and mild cases will increase over the following decade.

4.3. Scaling and Cost Reduction Pathways

Possible pathways for cost reduction include automation of cell culture and scaffold preparation, development of allogeneic cell lines to reduce per-patient costs, optimized production methods for growth factors, and economies of scale as production volumes increase. The treatment's cost must be evaluated against the lifetime cost of managing DCM with a cost range of $100,000-$1.6 million.[40,41] depending on severity. Quality-adjusted life years (QALYs) gained would be significant if the treatment prevents heart failure progression. If successful, the treatment could transform DCM from a chronic, progressive condition to a potentially curable disease. Given the high manufacturing costs but potentially transformative clinical benefits, this treatment would likely enter the market at a premium price of $50,000-100,000,[42,43] similar to other advanced cellular therapies.

5. Manufacturing

The manufacturability of this scaffold is very feasible. Figure 6 below summarizes the process, and the following sections elaborate on each stage in detail.

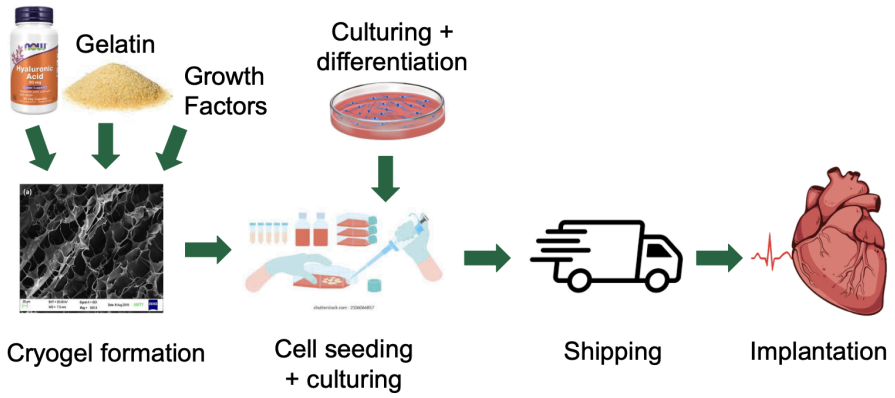


Figure 6. The manufacturing process of the proposed cryogel scaffold. First, the cryogel would be mixed and then frozen. Separately, cells would be bought, cultured, and differentiated in our lab. Second, these cells would be seeded in the cryogel scaffold, properly integrated, and cultured in an incubator. Then the scaffold would be shipped in a controlled environment to the hospital, where it would be reinserted in an incubator until it was ready for use in a patient.

5.1 Cryogelation

For cryogelation, hyaluronic acid, gelatin, and growth factors (EGF, VEGF, Periostin) will be mixed in an ARV-5000 THINKY mixer designed to mass mix hydrogels. These machines can mix up to 5kg of material and cost around $10,000[44]. The newly formed hydrogel would be moved into a large-scale Thermo-scientific freezer that is 51 cubic fit.[45,46] This freezer can tightly control temperatures to ensure consistent batch-to-batch porosity. Once extracted, these cryogels would be cut to the desired size.

5.2 Cell Culturing & Differentiation

The induced pluripotent stem cells (iPSCs) and mesenchymal stromal cells (MSC) that are intended to go in the scaffold will initially be bought from Reprocell and cultured and differentiated in vitro since they each need certain specific conditions to differentiate. iPSC cells will become cardiomyocytes by using a stirred-based bioreactor.[47] iPSCs will differentiate into cardiac progenitor cells (CPCs) by treating them with Glycogen Synthase Kinase (GSK) inhibitor "to induce the β-Catenin pathway which drives iPSC conversion to cardiac progenitors.[48,49] Finally, MSCs will differentiate into endothelial cells through a perfusion bioreactor with controlled shear stress. Each of these large-scale bioreactors can be estimated to cost around $50,000.[50]

5.3 Cell Seeding & Culturing

These differentiated cells will be seeded into the cryogel scaffold by using a perfusion bioreactor system that cultivates an oxygen-controlled environment.[51] This bioreactor has independent-operating bioreactors for large-scale production and is designed to control the environment and to make the method repeatable.[51] Once the cells are fully and properly integrated, they will go into a Fischer Scientific large-scale incubator for storage until shipping.

5.4 Shipping

Cells will be shipped in an EnviroTainer, a transport container that can control the environment to keep cells alive and the scaffold intact. These each cost $10,000; to start, 10 containers will be sufficient, so the total cost of these will be about $100,000.[52] Once the scaffolds reach the hospital, they will be stored in their incubator until they are ready for use.

5.5 Manufacturing Risks

During production, exposure to crosslinkers, initiators, and other chemical reagents can pose health risks to workers. Strict safety protocols, including the use of personal protective equipment (PPE) and controlled ventilation systems, must be implemented to ensure a safe working environment. The hazard mainly focuses on three aspects: chemical agent, cell components, and potential physical injuries.

The manufacturing process involves exposure to several chemical reagents and cell components, which can pose health risks to workers. To ensure a safe working environment, strict safety protocols must be implemented. The primary hazards in the manufacturing process can be categorized into three aspects. Firstly, chemical hazards arise from exposure to agents such as the crosslinker Methylenebisacrylamide, which can lead to skin irritation, respiratory issues, or other health concerns if not properly managed.[53] Secondly, biological hazards are associated with handling cell components, particularly the transgene agents used during the culturing of iPSCs. Lastly, physical hazards include the risk of injury from equipment handling, repetitive motions, and exposure to extreme temperatures during freezing and thawing cycles during the fabrication of cryogel. By addressing these risk areas through comprehensive safety measures and worker training, we aim to maintain a safe and compliant manufacturing environment.

5.6 Safety Controls

To address potential hazards and ensure the safety of our workers, we will implement a comprehensive safety control on both engineering and administrative level.

Our facility will be designed with dedicated sections that separate high-risk areas, such as storage zones for flammable materials and sterile environments within workshops and laboratories. These areas will be equipped with controlled ventilation systems, secure storage units, and emergency response equipment to minimize exposure risks and maintain strict environmental standards.

From an administrative perspective, we will provide thorough training programs for all staff, focusing on proper handling procedures for chemical reagents, biological materials, and equipment. Safety inspections for equipment will be conducted on a timely basis. Besides, safety regulations will be established to ensure consistent compliance across all operations.

What's more, appropriate personal protective equipment will be supplied to staff. This includes protective clothing, gloves, masks, and eye protection, tailored to the specific tasks and hazards present in each area of the facility. By integrating these safety controls, we aim to create a secure working environment that minimizes risks and promotes the health and well-being of our workforce.

6. Healthcare Economic Impact

6.1. United States Impact

Current dilated cardiomyopathy (DCM) treatments include medications priced at $5,000-10,000/year, implantable devices at $25,000-80,000, LVADs at $200,000, and heart transplants at $1.4-1.6 million.[38,39] Annual DCM-related hospitalizations cost on average $7 billion in the US.[54] A tissue engineering treatment designed as a long-term solution could reduce DCM treatment costs by potentially eliminating the need for transplantation in some patients and reducing hospitalizations.

Approximately 750,000[55,56] Americans have DCM, with annual new diagnoses of 35,000-40,000[55,56] patients. This represents a potential US market of $3.5-10 billion annually,[55,56] depending on adoption rate and pricing. This treatment would likely qualify for Medicare/Medicaid[57] coverage as a novel therapeutic for heart failure. Private insurers would require substantial clinical evidence showing superiority to current treatments, and procedural reimbursement codes would need to be established.

6.2. Global Impact

Developed markets would have similar economic considerations to the US market. Universal healthcare systems would require robust health technology assessments demonstrating cost-effectiveness. This treatment would be prohibitively expensive in emerging markets without significant cost reductions. There is potential for tiered pricing strategies based on country GDP, and technology transfer arrangements could enable regional manufacturing. Heart failure affects approximately 64 million people worldwide,[39] with DCM responsible for approximately 30-40% of all heart failure cases. The economic burden of heart failure globally exceeds $100 billion annually.[58]

7. Ethics

7.1. Societal Ethical Considerations

Introducing high-cost therapies requires ethical allocation frameworks that consider cost-effectiveness thresholds, quality-adjusted life-year calculations, opportunity costs, and potential age-based considerations. Insurance companies and patients' advocacy groups will be advised to work together to decrease these biases and make the treatment accessible. In addition, the development pathway raises research ethics issues regarding animal testing requirements, early human trial risks, proprietary technology limitations, and the balance between commercial interests and data transparency.

7.2. Clinical Practice Ethics

Clinicians face several ethical challenges, including determining appropriate patient selection criteria beyond clinical trial protocols. In addition, managing conflicts of interest if involved in both research and clinical care, balancing obligations to individual patients against stewardship of healthcare resources, and navigating learning curves that may affect early patient outcomes are crucial to creating extended ethical obligations for long-term monitoring compliant to FDA clinical stages and post-surveillance requirements.[42] International access considerations should also be taken into account.

7.3. Public Acceptance & Education

Public perception of biomaterials and regenerative medicine plays an important role in the commercial success of such innovative therapies. Public engagement through social media advertising and educational programs can help us address misconceptions and provide accurate information to the public.

To raise awareness of the social and political implications of engineering, scaffolds, and stem cell therapeutics, we plan to establish a presence on platforms like Instagram and TikTok. This will allow us to showcase our products, share educational content, and feature testimonials from patients who have benefited from our innovations.

Apart from online outreach, we also intend to organize educational lectures and workshops at middle schools, high schools, and colleges. These programs will introduce students to the field of tissue engineering and improve their understanding of how these technologies improve patients' quality of life.

By inspiring the next generation of scientists and promoting a broader understanding of regenerative medicine, we hope to contribute to a more informed and supportive public perspective.

7.4. Collaboration with Clinical Centers

Partnering with hospitals and medical institutions is important in facilitating preclinical and clinical trials, ultimately accelerating the regulatory approval process. Establishing strong interdisciplinary collaborations not only enhances product validation but also helps optimize protocol developments.

In addition, as previously mentioned, close collaboration with medical institutions will enable us to explore and implement more advanced shipping and distribution strategies. These partnerships can help reduce energy consumption, improve scaffold availability, and ensure timely treatment preparations.

7.5. Regulatory Challenges & Policy Advocacy

Securing approval from regulatory bodies such as the FDA and EMA requires strict adherence to rigorous safety, quality, and efficacy standards. Engaging in policy discussions and contributing to the development of industry guidelines can help smooth the pathway to market approval and integration into standard medical practices. Besides, from a strategic standpoint, we plan to introduce our acellular version of the scaffold as our initial product. Due to its relatively simpler regulatory pathway with the FDA, this approach will allow us to gain early market entry, validate our manufacturing processes, and establish credibility while preparing for more complex products that involve stem cell components.

8. Environmental Concerns

8.1. Material Sourcing & Sustainability

The cryogel production relies on biopolymers, crosslinkers, and bioactive agents. Sustainably sourcing these materials is crucial to minimize environmental impact and ensure long-term supply chain stability.

In our scaffold design, the primary materials are hyaluronic acid (HA) and gelatin, both of which are naturally derived. While these biopolymers can be produced through microbial fermentation, large-scale sourcing still often depends on animal-derived sources. This raises concerns regarding ethical sourcing, potential contamination risks, and supply chain reliability.[59] Animal farming practices also often involve the use of antibiotics, which can pose risks to the safety and quality of animal-derived materials such as FBS.[60] There are also challenges in maintaining consistent quality due to variations in natural sources.

Exploring plant-based or synthetic alternatives with comparable properties could significantly enhance sustainability. Switching to synthetic materials not only reduces reliance on animal products, contributing to improved animal welfare, but also lowers energy consumption and reduces the overall carbon footprint. This is largely due to the higher energy efficiency and lower carbon emissions associated with microbial fermentation compared to traditional animal farming.[61] Likewise, for our large-scale production, the high cost of natural materials also raises the need to shift toward more cost-effective synthetic options. In particular, like the large-scale use of HA in our scaffold. To address this, we are considering establishing an in-house production line for HA via bacterial fermentation. This approach would not only reduce manufacturing costs but also enhance control over product quality and strengthen supply chain stability.

8.2. Waste Management & Disposal

During the manufacturing processes various forms of waste could be generated, particularly chemical waste from the use of organic solvents and crosslinking agents. Effective waste management is essential to minimize environmental impact and ensure compliance with safety regulations.

The primary raw materials in our scaffold, including hyaluronic acid, gelatin, and components used in cell culture, are biodegradable and generally considered environmentally friendly. However, to further reduce material waste, we aim to optimize batch sizes and prevent stock degradation through careful inventory management. Apart from that, the cultured cells require strict disposal procedures in accordance with biosafety regulations. According to WHO laboratory biosafety manual, this involves inactivation followed by autoclaving before final disposal to prevent any potential biohazard risks.[62] Additionally, inorganic nano components like Ti_2C have potential bioaccumulation risk in aquatic environments.[63] To address this, any wastewater generated during scaffold fabrication will undergo proper filtration and chemical neutralization before disposal, ensuring compliance with environmental protection standards.

For waste generated during copolymerization, excess polymer solutions will be collected, solidified, and treated as biomedical waste. Rather than immediately disposing of failed production batches, we will consider repurposing defective cryogels for material property testing or non-clinical research purposes.

In terms of storage and shipping, we are committed to using recyclable or biodegradable materials for product packaging, further supporting our sustainability goals.

8.3. Energy Efficiency in Manufacturing

For future large-scale production, we aim to construct a factory based on green and environmentally friendly principles. This includes integrating clean energy sources, such as solar panels, and implementing advanced ventilation systems to naturally regulate indoor temperatures. This approach is inspired by sustainable designs like Apple Park in California, which utilizes on-site solar power that can generate 17 megawatts of power, sufficient to power 75% during peak daytime.[64]

Figure 7. Shipping components separately permits the use of existing cold chain systems. This more sustainable option allows us to capitalize on existing infrastructure, mitigating the need for energy-intensive, low-temperature conditions.

Regarding distribution, we propose a modular shipping strategy to further improve energy efficiency. The scaffold and cellular components could be shipped separately and assembled at clinical institutions prior to surgery. This approach reduces reliance on complex cold chain systems and lessens regulatory requirements for transporting sensitive cell components, as the scaffold can be preserved under standard cold chain conditions without compromising its functionality. Successful implementation of this strategy will require strong partnerships with medical centers, where trained personnel can assist with the final assembly and preparation of the product before clinical application.

Additionally, iPSCs or analogous MSCs can be harvested and cultured directly within clinical institutions and then assembled into our scaffolds. In this way, we can further reduce shipping demands, lower logistical costs, and enhance the accessibility of the treatment process.

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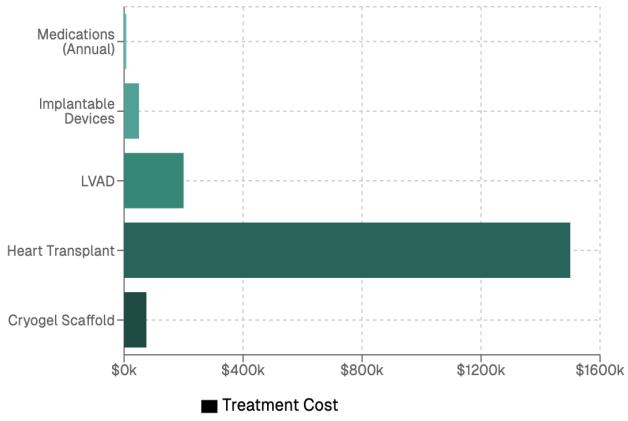
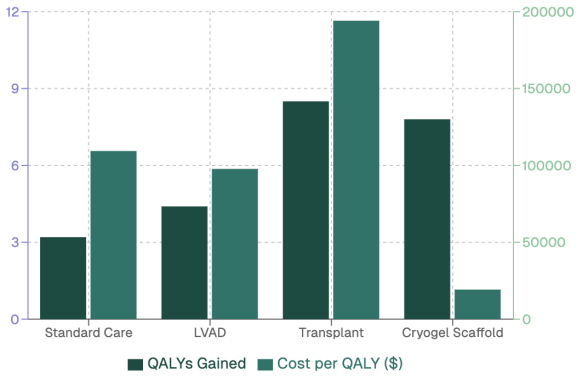
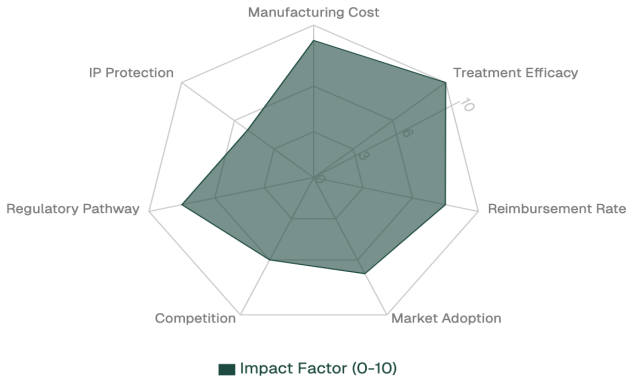
Appendix

Cost Comparison with Existing Treatments



QALY Comparison



Sensitivity Analysis